

physics modelling agent

update, Arxiv and units

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outline

1 arxiv problem

2 units

the arxiv issue

[c]

the problem

- arxiv queries where rejected
- papers could not be read or downloaded

the underlying issue

- crewai's ArxivPaperTool does not implement a proper exponential back-off and waiting time for the arxiv request
- ⇒ Arxiv-API-server rejects requests and even blocks IP address temporary

solution

solution

- exponential back-off and proper waiting time (5s between request)
- split the crew in two:
 - ① arxiv request and paper downloading (all in one step)
 - ② extract infos from downloaded papers; build the model and implement it

further improvements

- from one set of arxiv papers, the second crew can be run in multiple configurations without a new arxiv-server request
- ⇒ drastically reduces the number of server requests
- ⇒ papers could be added from another source into the directory the second crew reads from

the key formula

with

$$M_y = \frac{\mu_B |e| \tau}{2\pi} m^* \alpha_R \left[\hat{z} \times \vec{E} \right]_y$$

- μ_B Bohr magneton in J/T
- e charge in C
- τ relaxation time in s
- m^* effective mass in kg
- α_R rashba coupling strength in (J m)
- $\vec{E}$ electric field in V/m = kgm/(As³)
- M_y magnetization expected in A/m

changing α_R units to m⁴/s would result in matching units